MIL-S-19500/142A
13 May 1964
SUPERSEDING
MIL-S-19500/142(SigC)
20 September 1960

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICES, SILICON, HIGH VOLTAGE RECTIFIER

TYPES 1N1731A, 1N1733A, AND 1N1734A

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, semiconductor diodes for use in high-voltage rectifier circuits.

1.2 Absolute maximum ratings at +25°C

Type	v_r	V_R	V_{RM}(wkg)	i_f	I_o 1/	i_f (surge) 1/120-sec; T_A 100°C
	v(pk)	Vdc	v(pk)	a	mAdc	a
1N1731A	1800	1500	1500	1.05	350	6.0
1N1733A	3600	3000	3000	1.05	350	6.0
1N1734A	6000	5000	5000	1.05	350	6.0

1/ For ambient temperature above 25°C derate 2.0 mA/°C.

TEMPERATURE RANGE (OPERATING): T_A = -65°C to +150°C

TEMPERATURE RANGE (NONOPERATING): T_A = -65°C to +200°C

BAROMETRIC PRESSURE REDUCED (ALTITUDE OPERATION) 15 mm Hg., min.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with Specification MIL-S-19500, and as specified herein.

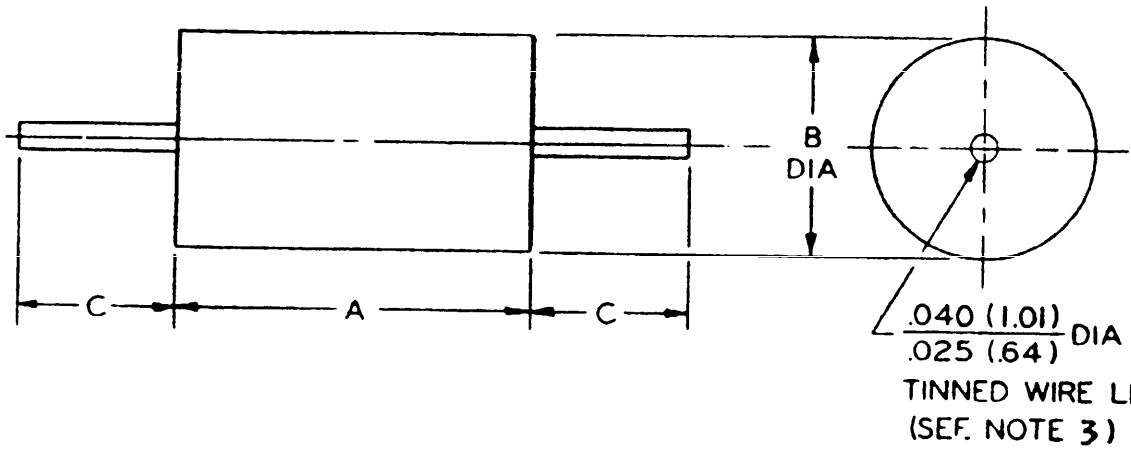
3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in Specification MIL-S-19500 and as follows:

V_{RM}(wkg)——— Working peak reverse voltage (maximum peak total value)

3.3 Design and construction. Semiconductor diodes shall be of the design, construction, and physical dimensions specified in figure 1.

FSC 5960

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TYPE	LTR	DIMENSIONS			
		INCHES		MILLIMETERS	
		MIN	MAX	MIN	MAX
1N1731A	A	.470	.530	11.94	13.46
	B	.345	.405	8.77	10.28
	C	1.250	1.375	31.75	34.92
1N1733A	A	.970	1.030	24.64	26.16
	B	.345	.405	8.77	10.28
	C	1.250	1.375	31.75	34.92
1N1734A	A	.970	1.030	24.64	26.16
	B	.345	.530	8.77	13.46
	C	1.250	1.375	31.75	34.92

NOTES:
1. Dimensions are in inches.
2. Metric dimensions are shown for general information only and are based upon 1 inch = 25.4 mm.
3. Gold plated leads may be substituted when specified in contract or order providing gold plated units conform to Table II Subgroup 5.

Figure 1. Semiconductor devices, diode, types 1N1731A, 1N1733A, 1N1734A.

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3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking shall be placed on the device:

(a) JAN prefix.

(b) Qualification designating code.

(c) Type designation.

(d) Acceptance date and inspection lot identification.

(e) Polarity marking (contrasting color band to indicate cathode end).

(f) Graphic polarity symbol.

3.5.1 The manufacturer at his option may omit the following markings:

(a) Manufacturer's identification.

(b) Country of origin.

3.5.2 All of the above markings, except polarity, shall be placed on the unit package.

3.5.3 Salt spray (corrosion). The devices shall be examined for destructive corrosion and illegible marking.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with Specification MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Acceptance inspection. Acceptance inspection shall consist of groups A, B, and C.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I, and shall be on a sublot basis.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

4.4.1 Inspection conditions. Unless otherwise specified herein all inspections shall be made at temperature (T_A) of 25°C.

4.4.2 Duty cycle requirements. This test shall be conducted with a half-sine wave at the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine wave form of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 130°; and the power shall be equal to or greater than that of a half-sine wave.

4.4.3 Pulse technique. The pulse width and duty cycle shall be selected so that doubling or halving the pulse width shall produce a minimum change in measured value. This is done to avoid errors caused by device heating during measurement of static or dynamic parameters.

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Table I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Min Rej No.	Symbol	Limits		
	Method	Details				Min	Max	Unit
Subgroup 1			10	5				
Visual and mechanical examination	2071							
Subgroup 2			5	4				
Forward voltage	4011	Pulse (see 4.4.3)			V_F			
1N1731A		I_F = 400 mAdc				0.8	3	vdc
1N1733A		I_F = 400 mAdc				2.4	6	vdc
1N1734A		I_F = 400 mAdc				4.0	8	vdc
Reverse current	4016				I_R			
1N1731A		V_R = 1500 Vdc				---	1	uAdc
1N1733A		V_R = 3000 Vdc				---	1	uAdc
1N1734A		V_R = 5000 Vdc				---	1	uAdc
Reverse current at peak reverse voltage	4016				i_r			
1N1731A		v_r = 1800 v(peak)				---	100	us(pk)
1N1733A		v_r = 3600 v(peak)				---	100	us(pk)
1N1734A		v_r = 6000 v(peak)				---	100	us(pk)
Reverse current 1/	4016	T_A = 150°C min			I_R			
1N1731A		V_R = 1500 Vdc				---	12	uAdc
1N1733A		V_R = 3000 Vdc				---	12	uAdc
1N1734A		V_R = 5000 Vdc				---	12	uAdc

1/ Measurement shall be made after thermal equilibrium has been reached at the temperature specified.

Table II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Min Rej No.	Symbol	Limits		
	Method	Details				Min	Max	Unit
Subgroup 1			15	7				
Physical dimensions	2066	See figure 1				---	---	---
Subgroup 2 1/			10	5				
Solderability	2026					---	---	---
Temperature cycling	1051	Test cond. C; 10 cycles				---	---	---
Thermal shock (glass strain)	1056	Test cond.B				---	---	---
Moisture resistance	1021	Omit initial conditioning				---	---	---
End points: Forward voltage	4011	Pulse (see 4.4.3)			V_F			
1N1731A		I_F = 400 mAdc				0.8	3	vdc
1N1733A		I_F = 400 mAdc				2.4	6	vdc
1N1734A		I_F = 400 mAdc				4.0	8	vdc
Reverse current	4016				I_R			
1N1731A		V_R = 1500 Vdc				---	2	uAdc
1N1733A		V_R = 3000 Vdc				---	2	uAdc
1N1734A		V_R = 5000 Vdc				---	2	uAdc

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Table II. Group B inspection. - Continued.

Examination or test	MIL-STD-750		LTPD	Min Rej No.	Symbol	Limits		
	Method	Details				Min	Max	Unit
Subgroup 3			10	5		---	---	---
Shock	2016	Nonoperating; 500 G peak; t = 1 msec 3 shocks each orientation X_1, Y_1; total 6 shocks						
Vibration fatigue	2046	Nonoperating; 10 G; orientations X_1, Y_1 only				---	---	---
Vibration, variable frequency	2056	Nonoperating; 20 G; orientations X_1, Y_1 only				---	---	---
Constant acceleration	2006	Nonoperating; 10,000 G each orientation X_1, Y_1				---	---	---
End points: (Same as subgroup 2, except I_R = 1 uAdc max)								
Subgroup 4 1/			10	5				
Terminal strength (lead fatigue)	2036	Test cond E				---	---	---
End points: (Same as subgroup 2)								
Subgroup 5 1/			10	5				
Salt spray (corrosion)	1046	Test cond B; (see 3.5.3)				---	---	---
End points: (Same as subgroup 2)								
Subgroup 6			10	5				
Surge current	4066	T_A = 100°C Ten 1/120-sec surges; 1 surge/minute superimposed on I_o = 200 mAdc i_f (surge) = 6a				---	---	---
End points: (Same as subgroup 2)								
Subgroup 7 2/			λ =10	5				
Steady state operation life	1026	(See 4.4.2) f = 60 cps				---	---	---
1N1731A		V_r = 1500 v(peak) I_o = 350 mAdc						
1N1733A		V_r = 3000 v(peak) I_o = 350 mAdc						
1N1734A		V_r = 5000 v(peak) I_o = 350 mAdc						
End points: (Same as subgroup 2)								

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greater than the value specified in 3.2.

Table II. Group B inspection. - Continued.

Examination or test	MIL-STD-750		LTPD	Min Rej No.	Symbol	Limits		
	Method	Details				Min	Max	Unit
Subgroup 8 High temperature life (nonoperating) End points: (Same as subgroup 2)	1031	T_A = 200°C	λ =10	5		---	---	---

1/ Destructive test.

2/ End point tests to be made at 340, 670 and 1000 hours.

Table III. Group C inspection.

Examination or test	MIL-STD-750		LTPD	Min Rej No.	Symbol	Limits		
	Method	Details				Min	Max	Unit
Subgroup 1 1/ Barometric pressure reduced (altitude operation) Measurement during test Reverse current 1N1731A 1N1733A 1N1734A	1001 4016	Pressure = 15 mm Hg t = 60 sec min. V_R = 1500 Vdc V_R = 3000 Vdc V_R = 5000 Vdc	10	3	 I_R	--- --- --- ---	--- 2 2 2	---- uAdc uAdc uAdc

1/ This test shall be conducted during qualification and thereafter every six months.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes. In addition to the note specified herein, the notes specified in Specification MIL-S-19500 are applicable to this specification.

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Navy - Ships
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Reviewers:
Army - EL
Navy - SH
Air Force - 11,17,85

Users:
Army - EL, MU, MI
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